

Combined central and peripheral demyelination: Clinical features, diagnostic findings, and treatment



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ARTICLE INFO

Article history:

Received 27 October 2015

Received in revised form 9 February 2016

Accepted 9 February 2016

Available online 10 February 2016

Keywords:

Combined central and peripheral demyelination

Chronic inflammatory demyelinating polyneuropathy

Multiple sclerosis

Acute disseminated encephalomyelitis

Guillain-Barré syndrome

ABSTRACT

Combined central and peripheral demyelination (CCPD) is rare, and current knowledge is based on case reports and small case series. The aim of our study was to describe the clinical features, diagnostic results, treatment and outcomes in a large cohort of patients with CCPD. Thirty-one patients entered this retrospective, observational, two-center study. In 20 patients (65%) CCPD presented, after an infection, as myeloradiculoneuropathy, encephalopathy, cranial neuropathy, length-dependent peripheral neuropathy, or pseudo-Guillain-Barré syndrome. Demyelinating features of peripheral nerve damage fulfilling European Federation of Neurological Societies/Peripheral Nerve Society (EFNS/PNS) electrodiagnostic criteria for CIDP were found in 23 patients (74%), and spatial dissemination of demyelinating lesions on brain MRI fulfilling the 2010 McDonald criteria for multiple sclerosis (MS) in 11 (46%). Two thirds of the patients had a relapsing or progressive disease course, usually related to the appearance of new spinal cord lesions or worsening of the peripheral neuropathy, and showed unsatisfactory responses to high-dose corticosteroids and intravenous immunoglobulins. The clinical presentation of CCPD was severe in 22 patients (71%), who were left significantly disabled. Our data suggest that CCPD has heterogeneous features and shows frequent post-infectious onset, primary peripheral nervous system or central nervous system involvement, a monophasic or chronic disease course, inadequate response to treatments, and a generally poor outcome. We therefore conclude that the current diagnostic criteria for MS and CIDP may not fully encompass the spectrum of possible manifestations of CCPD, whose pathogenesis remains largely unknown.

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1. Introduction

Inflammatory demyelinating diseases are a broad group of disorders characterised by immune-mediated myelin damage, leading to conduction abnormalities, and often accompanied by axonal loss. The demyelinating lesions are usually limited either to the central nervous system (CNS) or to the peripheral nervous system (PNS). The simultaneous occurrence of combined central and peripheral demyelination (CCPD) is rare and data are limited to case reports and small case series [1, 11, 13].

We previously found that 36% of a selected cohort of patients with post-infectious inflammatory demyelinating disorders of the CNS showed either clinical or electroneurographic evidence of PNS involvement.

Notably, these patients with CCPD showed a worse prognosis and a higher relapse rate compared to those with CNS-restricted variants [8].

The aim of this study was to examine the clinical and paraclinical features of a large cohort of patients with CCPD in order to gain information about clinical presentation, disease course, response to treatments and outcome.

2. Patients and methods

This is a retrospective, observational, two-centre study. The inclusion criteria for CCPD patient selection were: A) acute (within the previous month), subacute (1–2 months previously) or chronic (over 2 months previously) onset of symptoms of CNS and/or PNS impairment; B) presence of CNS lesions suggestive of demyelination shown by brain and/or spine MRI; C) presence of peripheral neuropathy shown by nerve conduction studies (NCS); and D) exclusion of other causes of combined CNS and PNS involvement, as detailed below.

DOI of original article: <http://dx.doi.org/10.1016/j.jns.2016.02.023>.

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Of 276 cases with idiopathic inflammatory diseases of the CNS or PNS who were identified from clinical databases and followed up at the C. Mondino and C. Besta Neurological Institutes, 31 patients fulfilled all the inclusion criteria and entered the study (Supplementary Fig. 1). The study was approved by the C-Mondino Institutional Review Board and written informed consent was obtained from all the subjects.

Clinical data were collected from inpatient and outpatient hospital charts. For descriptive purposes, disease course was classified into three categories: 1) monophasic, if characterised by a single episode of acute demyelination, not followed by any further events; 2) relapsing, if the first event was followed, at least 3 months after the initial episode, by a clinical relapse defined by acute/subacute onset of new symptoms, or recurrence of previously experienced symptoms; and 3) chronic progressive, defined by the presence of a steady or stepwise worsening of the symptoms. Original MRI scans (1.5 T), cerebrospinal fluid (CSF) examination, including agarose isoelectric focusing for oligoclonal IgG band (OCB) determination, and NCSs were reviewed for technical and interpretation accuracy. In the NCS analyses we did not include: a) nerve roots and peripheral nerves stemming from metameres affected by inflammatory lesions, as detected by spine MRI; b) nerves showing alterations at entrapment points, possibly related to prolonged immobility.

The following investigations were performed in all the patients in order to exclude other causes of combined CNS and PNS involvement, as well as distinct causes of neuropathy: fasting blood glucose, Hb1Ac, vitamin B12, folates, homocysteine, serum immunofixation, serology for HIV, HCV, Lyme disease, syphilis, ANA, ENA, ANCA, anti-cerebellum, anti AQP-4 antibodies (cell-based assay), anti-gangliosides, including anti GQ1b, and anti-sulfatide antibodies, serum ACE. In patients with a chronic progressive disease course, metabolic testing for phytanic acid, galactocerebrosidase, arylsulfatase, very long chain fatty acids, and genetic testing for mutations in *PMP22* (CMT1A) and *GJB1* (CMTX1) were also performed.

The clinical charts were reviewed and the following information was recorded for all the patients: demographics, general medical history, risk factors for neuropathy, symptoms at presentation, previous infections and vaccinations, outcome and response to treatment. The modified Rankin scale (mRS) was used to measure disability level and outcome. Responders were defined as patients recording an increase of at least one point on the mRS after treatment. The first available neurological examination from disease onset was recorded. Muscle weakness was graded as mild (from 5- to 4 of Medical Research Council (MRC) scale), moderate (from 4- to 3 of MRC scale) and severe/paralysis (<3).

Original NCSs were evaluated for fulfilment of European Federation of Neurological Societies/Peripheral Nerve Society (EFNS/PNS) electrodiagnostic criteria for chronic inflammatory demyelination polyradiculoneuropathy (CIDP) [4]. Original brain MRI scans were reviewed for fulfilment of the 2010 McDonald criteria for multiple sclerosis (MS) [10].

Data were analysed using descriptive statistics methods. Continuous data were shown as means or medians (standard deviation, min-max).

3. Results

3.1. Clinical presentation

Table 1 shows the demographic and clinical features at disease onset and at follow-up of 31 patients with CCPD. Each patient's disease history is summarised in Supplementary Table 1. The majority of the patients were male with a mean age at onset of 57 years (range, 14–82). A previous infection or vaccination was reported in 20 subjects (65%), mostly infections of the upper respiratory tract, although a specific trigger was identified in only four cases (1 flu vaccine, 1 *Streptococcus pneumoniae* vaccine, 1 *Campylobacter jejuni* gastroenteritis and 1 *Streptococcus pyogenes* pharyngitis). The majority of the patients presented with lower limb sensory-motor impairment and sphincter dysfunction,

Table 1

Clinical features of CCPD patients at disease onset and in the chronic phase of the disease.

Demographics		
Age at onset in years (standard deviation, range)	57 (17, 14–82)	
Males	23 (74%)	
Disease course		
Monophasic	10 (32%)	
Relapsing	13 (42%)	
Chronic progressive	8 (26%) ^b	
	Disease onset	Follow-up ^a
Previous infection	20 (65%)	1 (5%)
Clinical features		
Lower limb sensory-motor impairment	29 (94%)	19 (90%)
Urinary incontinence/retention	26 (84%)	2 (9%)
Distal paresthesia	11 (35%)	4 (19%)
Altered mental status	5 (16%)	–
Upper limb sensory-motor impairment	8 (26%)	7 (33%)
Headache	2 (6%)	–
Ascending four limb sensory-motor impairment	4 (13%)	–
Other	7 (23%)	4 (19%)
Response to treatments		
Steroids	17/23 (74%)	4/16 (25%)
IVIg	4/8 (50%)	4/11 (36%)
Other	1/1 (100%)	1/4 (25%)
Overall response	19 (73%)	6 (19%)
Disability (mRS)		
N = 25 ^c		N = 31
0	1 (4%)	2 (7%)
1	1 (4%)	2 (7%)
2	1 (4%)	1 (3%)
3	6 (24%)	4 (13%)
4	12 (48%)	4 (13%)
5	4 (16%)	18 (58%)

All data are reported as number of cases (%) or mean (standard deviation, min-max). CCPD: combined central and peripheral demyelination; IVIg: intravenous immunoglobulins; mRS: modified Rankin Scale.

^a Follow-up data for previous infection and clinical features refer only to cases with a relapsing or chronic progressive disease course (N = 21).

^b Including 6 cases with a primary progressive disease course.

^c Primary progressive CCPD cases are excluded.

suggesting the presence of spinal cord lesions. However other symptoms, such as altered mental status and cranial nerve involvement, could also characterise the onset of the disease. Primary PNS damage was present in 11 patients who presented with distal paraesthesia, and in four subjects with pseudo-Guillain-Barré syndrome, i.e. with ascending sensory-motor impairment and abolished osteotendinous reflexes.

3.2. Neurological examination

In 27 patients (87%), neurological examination at disease peak showed lower limb weakness which was severe in 67% of them, with distal predominance in five. Upper limb weakness was found in 12 patients (39%). Tone was increased at the four limbs in 11 patients, decreased at the four limbs in six, and mixed (increased at the upper limbs and decreased at the lower limbs) in two. Osteotendinous reflexes were more often increased at the upper limbs (48% of the cases) and reduced or abolished at the lower limbs (52%). Babinski sign was present in 58% of the cases. Dystonia was observed in one case (Supplementary Table 2).

3.3. Disease course

The patients were followed up for a mean of 84.3 months (range, 43.3–134). One third of the patients showed a monophasic disease course. Overall, in 21 cases (68%) we observed a progression of the disease: either a relapse, with subacute onset of new symptoms (13 cases, 42%), or a steady chronic progression (8 cases, 26%). Notably, six patients who presented with distal paraesthesia showed a progressive disease course from onset. In the relapsing subgroup, the mean delay between inflammatory events was 12 months (50.3–126). In five

cases, the relapse occurred during the first year after steroid withdrawal. Clinically, relapses and chronic progressions were characterised by worsening of lower limb sensory-motor impairment, onset of upper limb paralysis, or brainstem dysfunction, but not encephalopathic symptoms.

3.4. Treatment and outcome

In the acute phase, 26 patients (84%) were treated with steroids, either 6-methyl-prednisolone 500–1000 mg/day for five days followed by oral tapering over three months, or chronic prednisone 1 mg/kg/day, intravenous immunoglobulins (IVIg) or plasma exchange (Table 1). An mRS score improvement of at least one point was observed in 19/26 (73%). In 24 patients with relapses or chronic progression either steroids or IVIg were used later in the disease course, but the response rate was lower (19%) than in those in whom they were used at disease onset. Azathioprine and methotrexate were used in just two cases with a chronic progressive disease course, and did not produce a significant improvement.

Of interest, natalizumab was used as an off-label drug in a young patient who was unresponsive to steroids and showed persistent inflammatory activity on brain MRI. The drug strikingly improved the brain MRI lesions. However, the patient's clinical condition deteriorated due to a marked worsening of nerve conduction, and the treatment was therefore discontinued. The patient was subsequently started on rituximab with persistent clinical benefit (follow-up, 12 months).

Overall, the outcome was poor (mRS ≥ 4) in 22 cases (71%).

3.5. Cerebrospinal fluid examination

CSF analysis showed pleocytosis and elevated total proteins in 58% and 74% of the patients, respectively (Table 2). Fourteen of the 31 cases (45%) showed intrathecal IgG production, which was detected as either CSF-restricted OCBs (5 cases), or as the presence of OCBs that were equal in serum and CSF in addition to CSF-restricted OCBs (9 cases). CSF analysis was repeated in seven patients with a chronic disease course. In the only two patients with CSF-restricted OCBs at onset, the OCB pattern changed: the OCBs disappeared in one case and changed to a mirror pattern, namely absence of intrathecal IgG production, in the other. In the remaining five patients, CSF analysis did not show OCBs at either disease onset or later in the disease course.

3.6. MRI

The CNS involvement in our cohort was characterised by multifocal demyelinating lesions of the brain, and focal, multifocal and, less frequently, longitudinally extensive lesions involving the posterior-lateral white matter of the spinal cord (Table 2, Fig. 1A–I, Supplementary Figs. 2–5). Half of the patients had active lesions, mostly in the spinal cord, and 17 (56.7%) showed contrast enhancement of the cauda and nerve roots, which tended to persist over time. Eleven patients (46%) fulfilled the 2010 McDonald criteria for MS for dissemination in space [10]. On subsequent MRI scans, temporal dissemination of lesions occurred in 7 patients (22.6%) who had either a relapsing or a progressive disease course, but not in patients with a monophasic disease course. New lesions occurred more frequently in the spinal cord.

3.7. Evoked potentials

Somatosensory evoked potentials from the upper limbs were abnormal in 10/18 subjects (55%), whereas those from the lower limbs were abnormal in all the subjects – either slowed (in six patients) or not elicitable (in 12) – due to abnormal peripheral nerve conduction. Visual evoked potentials (VEPs) were altered in 9/14 patients (64%) and brainstem auditory evoked potentials in 5/13 patients (38%).

Table 2
MRI and CSF findings in CCPD.

Brain MRI	N = 26
T2 lesions	19/26 (73%)
Topography	
Periventricular	12/19 (63%)
Subcortical	12/19 (63%)
Infratentorial	10/19 (53%)
Deep grey matter	2/19 (10.5%)
Type of lesion	
MS-like	10/19 (53%)
ADEM-like	2/19 (10%)
Non-specific	7/19 (37%)
T1 lesions with Gd enhancement	3/26 (11%)
Spine MRI	N = 30
T2 lesions	24/30 (80%)
Topography	
Posterolateral white matter	16/24 (66.7%)
Anterior white matter	3/24 (12.5%)
Central grey matter	3/24 (12.5%)
White and grey matter	2/24 (8.3%)
Distribution	
Cervical	14/24 (58%)
Thoracic	16/24 (67%)
Lumbosacral (conus-epiconus)	9/24 (37%)
Type of lesion	
Focal	9/24 (37%)
Multifocal	9/24 (37%)
LETM	5/24 (21%)
T1 lesions with Gd enhancement	15/30 (50%)
T1 meningeal, nerve root and conus/cauda Gd enhancement	17/30 (56.7%)
Fulfilling the 2010 McDonald criteria for MS for dissemination in space	12/26 (46%)
Temporal dissemination	7/31 (22.6%)
CSF examination	N = 31
Pleocytosis	18 (58%), 54 (32, 8–144)
Raised proteins	23 (74%), 97 (44, 22–200)
OCB	
CSF-restricted OCB	5/31 (16%)
OCB equal in serum and CSF plus CSF-restricted OCB	9/31 (29%)

All data are reported as number of cases (%), except pleocytosis and raised CSF proteins, which are reported as number of cases (%), median (standard deviation, min-max). CSF pleocytosis is defined as more than 5 cells/ μ L; raised CSF proteins are defined as more than 40 mg/dL. ADEM: acute demyelinating encephalomyelitis, Gd: gadolinium, LETM: longitudinally extensive transverse myelitis. MS: multiple sclerosis, OCB: oligoclonal IgG bands.

3.8. Nerve conduction study

Twenty-three patients (74%) fulfilled the EFNS/PNS electrodiagnostic criteria for CIDP [4]: definite in 14 (45%), probable in three (10%) and possible in six (19%), while eight cases (26%) had pure axonal neuropathy. Demyelination showed a proximal-distal gradient as documented by frequent F-response latency prolongation (6 cases, 19%) or absence (7 cases, 23%), and slow nerve conduction velocities (13 cases, 42%), as opposed to relatively preserved distal motor latencies (2 cases, 6%) (Fig. 1–J). Complete and partial conduction blocks were present in five (16%) and eight (26%) patients, respectively.

Peroneal and tibial nerves were the most frequently affected motor nerves, showing mixed demyelinating and axonal damage. Median and ulnar nerves often showed demyelinating involvement with relatively preserved motor amplitudes. Sensory nerve conduction was altered at the sural nerves and, to a lesser extent, at the median nerves (Table 3).

4. Discussion

The clinical and paraclinical data in the present cohort of patients, in keeping with previous reports [1, 6, 9, 11, 13], show that CCPD is a rather heterogeneous disease.

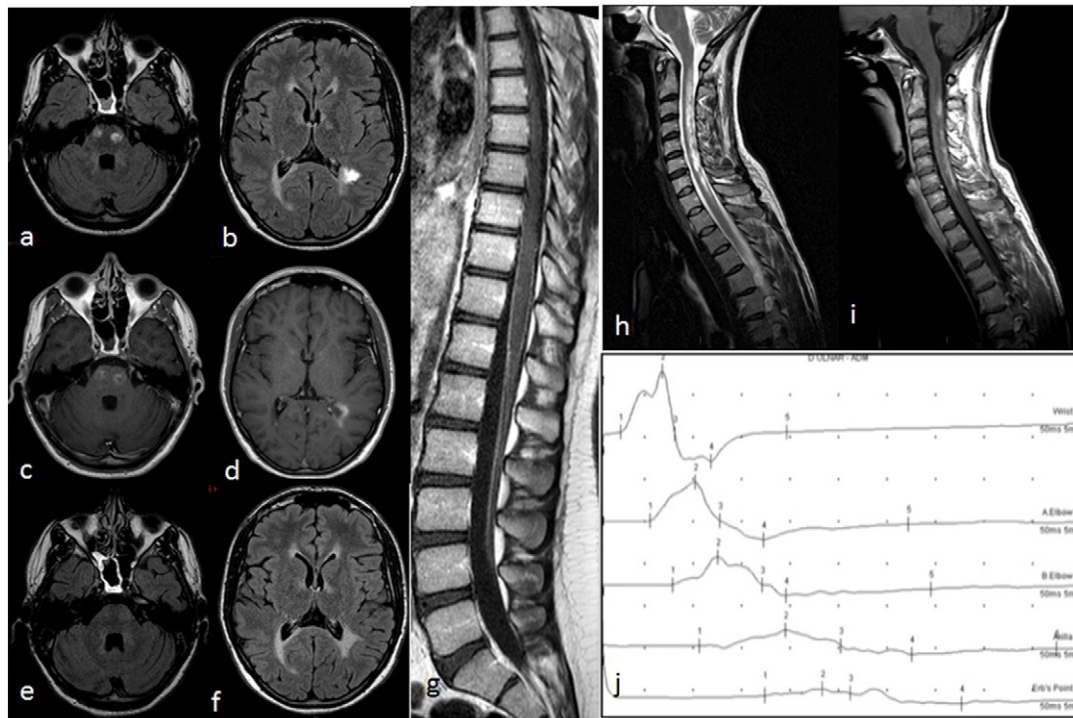


Fig. 1. MRI and nerve conduction study findings in CCPD. Brain axial FLAIR images (a, b, e, f) and T1-w images post-gadolinium (gd) administration (c, d) performed pre- (a, b, c, d) and post- (e, f) immunosuppressive treatment with rituximab in a patient with CCPD. Pre-treatment spine MRI with T1-w image post-gd (g). Pre-treatment brain MRI (a, b, c, d) highlights, in the context of diffuse, mainly periventricular, lesions, three round/ovoid hyperintense signal alterations, at the level of the pons (a) and in the left periventricular region (b) showing gd enhancement (c, d). Post-treatment brain MRI shows marked reduction of the pons and left periventricular lesions (e, f). Pre-treatment spine MRI (g) shows marked enhancement of both caudal root and conus lesions. Spine MRI with sagittal T1-w image (h) and post-gd T1-w image (i) showing an extensive cervical spine lesion from C2 to D1 (h), hyperintense on T2- and isohypointense on T1-w images, where a patchy vivid enhancement is detectable (i), mainly posteriorly. (j) Motor nerve conduction study of the ulnar nerve (recording from muscle abductor digiti minimi) showing temporal dispersion of the compound muscle action potential (CMAP) in proximal segments (roots and plexuses) in a patient with CCPD. Calibration: 5 ms/div; 5 mV/div.

A documented infection preceded CCPD onset in 65% of cases, but was unusual before relapses. This percentage is higher than that reported recently in a Japanese CCPD cohort (10%) [9], but is consistent with a previous observation showing that the large majority (85%) of pediatric CCPD cases followed an infection [1]. The high frequency of infections before CCPD onset argues for a role of infectious agents as triggers of subsequent self-sustained autoimmune processes.

PNS and CNS involvement was simultaneous or rapidly subsequent in 22 cases (71%), CNS preceded PNS involvement in six (19%) cases, whereas PNS preceded CNS involvement in three (10%). However, in some patients, symptoms could not easily be attributed to either primary PNS or CNS involvement, and subclinical alterations limited to one compartment could have been present before the clinical onset.

Relapsing symptoms occurred in 42% of the cases, and were mainly due to new CNS lesions, while PNS symptoms mainly showed an insidious onset and persisted over time. The preponderance of cases with simultaneous PNS and CNS demyelination refutes the idea of a spreading of autoimmunity from one compartment to the other, following exposure to previously hidden epitopes. Conversely, it is conceivable that autoreactive lymphocytes and/or antibodies against a common PNS and CNS epitope may cross the blood-brain and blood-nerve barriers and activate an inflammatory process concomitantly, as previously suggested [5].

With regard to the PNS damage, NCS disclosed clear demyelinating features in approximately 3/4 of the patients, with conduction blocks in 42% of them, while the other 26% had a pure axonal neuropathy.

Table 3
Nerve conduction study in patients with CCPD.

	Motor						Sensory		
	Altered/tested	CMAP (mV)	MCV (m/s)	DML (ms)	F wave latency (ms)		Altered	SAP (μV)	SCV (m/s)
Ref		>2	>40.6	<5.8		Ref		>6	>42
Peroneus R	17/23	0.9 (0.9)	36.1 (5.4)	4.3 (0.8)		Suralis	14/23	3.1 (2.3)	38 (1.9)
Peroneus L	16/22	0.5 (0.5)	35.1 (4.4)	4.5 (1)			15/23	2.5 (1.6)	37.2 (2.1)
Ref		>5	>41	<5.5					
Tibialis R	19/26	0.5 (0.5)	35.1 (4.4)	4.7 (0.8)	70 (9.8)				
Tibialis L	17/26	2.2 (1.3)	33.3 (4.7)	4.6 (1)	67 (9.7)				
Ref		>5	>48	<3.3		Ref		>6	>46
Ulnar R	8/12	7.6 (2.9)	30.7 (8.3)	3 (0.6)	41.1 (10)	Ulnar	1/10	12 (12.5)	49.1 (6.1)
Tibialis L	7/12	12.8 (14)	36.7 (8.1)	3 (0.6)	39.8 (7.5)		1/10	13 (10.2)	44.6 (10.8)
Ref		>5	>46.8	<4		Ref		>8	>46.8
Median R	11/15	8.5 (4.3)	37.7 (9.5)	3.7 (0.5)	35.7 (5.9)	Median	5/8	3.7 (2.2)	26 (4.6)
Tibialis L	8/13	6.6 (6)	36.6 (5.6)	3.8 (0.9)	36.4 (6.2)		4/10	3.6 (4.6)	38 (2)

CMAP: compound motor action potential; MCV: motor conduction velocity; DML: distal motor latency; SAP: sensory action potential; SCV: sensory conduction velocity. Data are indicated as mean (standard deviation).

However, in these latter cases, we cannot exclude the possibility that demyelinating changes had initially been present and could not be detected in later disease stages due to severe axonal degeneration.

As regards the CNS damage, demyelinating lesions documented on MRI scans affected the brain and the spinal cord, particularly the dorsal and lumbosacral metamer, and were frequently associated with Gd enhancement of the cauda and nerve roots. Bilateral optic nerve involvement was often asymptomatic but could be detected on VEPs in 64% of tested cases. Of note, six anti-aquaporin-4 antibody-seronegative cases had spinal cord lesions extending over three metamer, which were thus indistinguishable from those typically seen in neuromyelitis optica spectrum disorders; this suggests that CCPD should be considered in the differential diagnosis of these disorders. Unfortunately, these patients' serum samples are unavailable for anti-MOG antibody testing.

Overall, 46% of the patients fulfilled the 2010 McDonald criteria for MS and 74% fulfilled the EFNS/PNS electrodiagnostic criteria for CIDP, independently of their clinical presentation, disease course, response to treatment and outcome (data not shown). For this reason, quite apart from the fact that MS and CIDP can co-occur, definition of CCPD based on current diagnostic criteria for MS and CIDP may be too restrictive and arguably exclude some patients with combined inflammatory disease of the CNS and PNS, in whom extensive clinical investigation is unable to provide a better explanation. Analysis of larger international case series of patients with combined inflammatory disorders of the CNS and PNS will be useful in order to define the spectrum of "typical" CCPD and its variants.

Of interest, 2/3 of the patients entered the chronic phase of the disease, with either a relapsing or a progressive disease course, and similarly to what is seen in MS. However, the features that differentiate CCPD from MS in our series are: 1) the OCB patterns: persistent CSF-restricted OCBs, which are typical of MS, were found in only two patients, whereas OCBs that were equal in serum and CSF in addition to CSF-restricted OCBs were more frequently detected. This latter abnormality suggests the presence of mixed systemic plus intrathecal humoral immune activation. Nonetheless, we cannot exclude the possibility that patients whose CSF was not tested repeatedly might have developed OCBs later in the disease course; 2) the lack of asymptomatic spatial dissemination of lesions on MRI; and 3) the coexistence with PNS involvement itself. The CCPD cases here reported also differ from typical acute demyelinating encephalomyelitis in their higher age at onset, the rarity of large confluent and synchronous lesions on brain MRI affecting both white and grey matter, the relapsing and progressive courses shown by the majority of our cases, and their overall poor outcome ([7] International Pediatric Multiple Sclerosis Study Group).

The efficacy of steroids and IVIg was only partial and mostly limited to the acute phase of the disease. Rituximab, whose efficacy has been observed in both MS and CIDP [2, 3], led to marked improvement in a CCPD patient with an aggressive disease course and may thus represent an option for the treatment of steroid- and IVIg-resistant cases. Conversely, natalizumab, although highly effective in dampening CNS inflammatory activity in one case with MS-like lesions on brain MRI, exacerbated PNS symptoms, possibly due to drug-induced inhibition of blood-brain barrier crossing by pathogenic T cells, which therefore concentrate systemically. A previous study also reported inefficacy of natalizumab in CIDP patients [12].

Clinically, the outcome was poor in 71% of our patients, which is consistent with the previous observation of an unfavourable outcome with Expanded Disability Status Scale score ≥ 5 in 54% of children with CCPD [1], but appears to be worse than what was reported in a previous study in which 26 out of 40 CCPD cases (65%) had no or mild disability (Hughes functional score ≤ 1) [9]. The better outcome of the Japanese patients [9] could be due to differences in demographics (a majority of young females in the Japanese series, as opposed to the prevalence of older males in our study) and paraclinical features, with the Japanese study showing a higher frequency of patients with definite demyelination on NCS and cerebral lesions fulfilling the 2010 McDonald criteria for MS.

In addition, the presence of different pathogenic mechanisms cannot be excluded.

Recently, antibodies to neurofascin-155 (NF155), a protein EXPRESSED on both central and peripheral myelin, and involved in axo-glial coupling at the paranodal regions which flank the node of Ranvier were identified in five out of seven Japanese patients with CCPD [6] who showed a relapsing disease course and, unlike our patients, a good response to IVIg. Also, antibodies to NF155 have been identified in a low percentage of patients affected by CIDP who had predominantly distal weakness, ataxia, disabling tremor and poor response to IVIg [14]. A possible cerebellar origin of tremor was hypothesised, although there was no conclusive evidence of CNS involvement in these patients. However, their clinical phenotype differed from that of our patients who invariably showed symptomatic CNS damage, mainly long tract impairment and altered mental status, but no tremor or cerebellar ataxia. Despite differences in clinical features, serological characterisation for NF155 antibodies in larger cohorts of CCPD patients may thus help to define subtypes of patients with different prognoses and different responses to treatment.

This study has some limitations. First, the retrospective design makes it difficult to define the efficacy of the treatments. Second, the clinical evaluations were assessed at different time points in the disease course, which may have resulted in non-homogeneous biological and instrumental findings. Third, serum samples of only a limited number of patients had been stored and are available for testing. For this reason we were not able to assess the presence of anti-NF155 antibodies in the entire cohort, or in a representative proportion of it.

In conclusion, our large cohort of CCPD patients was characterised by heterogeneous clinical manifestations, with either primary PNS or primary CNS involvement, frequent preceding infections, and a monophasic or chronic disease course. The outcome was poor in the majority of the patients and response to treatments, including steroids and IVIg, was partial and mostly limited to the acute phase of the disease. Definition of CCPD based on current diagnostic criteria for MS and CIDP may be too restrictive due to the broad spectrum of combined inflammatory damage of the CNS and PNS, whose pathogenesis remains largely unknown.

Supplementary data to this article can be found online at <http://dx.doi.org/10.1016/j.jns.2016.02.022>.

Study funding

This project was supported by the Italian Ministry of Health (120/RF-2011-02347955, EM).

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